

METAL at Work

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3 billion human lives ended on August 29th, 1997. The survivors of the nuclear fire called the war Judgment Day. They lived only to face a new nightmare, the war against the Machines...

A metal foot crushes the skull like china.

TILT UP, revealing a humanoid machine holding a massive battle rifle.

It is the endoskeleton of a Series 800 terminator. Its glowing red eyes compassionlessly sweep the dead terrain, hunting.

—Terminator 2: Judgement Day

This was the future. They all told us so—Asimov and Cameron—our home was going to be dominated by metalmen. And if we weren't careful, they would stop serving us coffee and toast, and take over the world. But the future is already here and instead of menacing cold wires, and exoskeletons with laser eyes, we have metal canines going into a frenzy over a plastic ball. Much as our soothsayers would like to make us believe, robots are used for more menial labour than world domination—tasks unglamorous enough to restrict their menacing frame to a giant limb with enough mobility to pick up and place components of a car frame firmly in place.

The beauty of robotics is that it brings together different disciplines such as mechanical engineering, electronics, digital electronics, communication, computer science, algorithms and chemical engineering. And it also diversifies into other

areas such as Artificial Intelligence, a prime driver for the future robotics and Hollywood movies alike.

Robots are more used in the industry scenario to increase precision or productivity. The humble industrial robot is a tool used to do routine and tedious jobs, where human accuracy suffers over a period of time. The typical working day for an industry robot involves welding, assembly, spray painting and replacing the nuclear fuel rods in nuclear power plants—jobs that are too monotonous or harmful for human beings. While industrial robots are a widespread norm in countries such as Japan, Russia, Germany, the United States and China, they haven't been able to catch up too quickly in India for reasons financial and human. Which is not to say that they aren't here at all—robots do thrive in the clinical environment of the mechanical laboratories of our engineering universities. But will they come out to butter our toast?

Still in school

C. Amarnath, teaching in the mechanical engineering department of IIT, Mumbai, has worked on the nine-year project, Natraj, along with Isaac Kurien and the students of mechanical engineering. The Natraj is a walking robot built using completely indigenous components. The development of the Natraj is funded by the Bhabha Atomic Research Centre

(BARC) which aims to use it in its nuclear plants for hazardous operations. So what's so special about it? Amarnath explains: "The Natraj is a walking machine and it is fairly more difficult to define its moves. For a robot to walk, we have to study the human sense of balance and movement very closely and try to emulate it in the mechanism. It has to learn to work with gravity and balance itself while moving its feet." Kurien agrees: "Although all of us walk, it's hard to define it for a machine. They don't have a sense of balance and tend to overturn."

Making a robot walk takes it to a higher level of mechanical intelligence and gives it mobility. This, in turn, makes it applicable to more areas of work. It can now navigate itself around to do the prescribed task. As Subir Saha, associate professor in IIT, Delhi, and an ardent champion of industrial robotics in India says, "In the Indian way of life, drudgery and life hazardous jobs are the norm. Anyone exposed to nuclear, radiation or working in steel factories faces occupational danger. In a scenario like this, there is no reason why robotics should not be applied for the welfare of the human workers."

And what is the future workday of the Natraj going to be like? According to Kurien, "With the Natraj, we've tried to prove, using indigenous technology, that machines can walk. The next step would be to make it more intelligent using

